

Chapter 5

Trigonometry

5.1 Radian measure and the circle

A radian measure is the angle subtended at the centre of a circle by an arc with length equal to that of the radius. It follows that θ radians is equal to $\frac{180}{\pi}\theta$, so that π radians = 180 degrees.

The following examples will help us familiarise with converting from degrees to radians and vice versa.

Example 5.1.1. Express each of the following as radians

1. 45°
2. -30°
3. 270°

Example 5.1.2. Express the following radian measures in degrees.

1. $\frac{5\pi}{6}$
2. $\frac{3\pi}{2}$
3. $\frac{-7\pi}{4}$

5.2 Definition of the trigonometric ratios

Suppose the arm OR of unit length in the first can rotate about O (the origin) in an anticlockwise direction and makes an angle θ with the positive x -axis. We divide the complete revolution (360°) into 4 quadrants and take the positive y -axis at 90° . Let (x, y) be the coordinates

of R . We note that x and y will be positive or negative depending on the quadrant R lies in.

If ORA , A a point on the x -axis, is the right-angled triangle, then the trigonometric ratios are defined as follows:

1. $\frac{OA}{OR} = \cos \theta$ (cosine of θ).
2. $\frac{AR}{OR} = \sin \theta$ (sine of θ).
3. $\frac{AR}{OA} = \tan \theta$ (tangent of θ).
4. $\frac{OR}{OA} = \sec \theta$ (secant of θ) = $\frac{1}{\cos \theta}$.
5. $\frac{OR}{AR} = \csc \theta$ (cosecant of θ) = $\frac{1}{\sin \theta}$.
6. $\frac{OA}{AR} = \cot \theta$ (cotangent of θ) = $\frac{1}{\tan \theta}$.

Treating θ as an independent variable, these might also be regarded as trigonometric functions of θ . For general angles, greater than 90° the sign of each ratio depends on the quadrant in which the angle lies. Since we shall be required to determine the trigonometric ratios of not only the acute angles, we shall use what we shall term an associated acute angle to determine ratios of an angle θ when it is greater than 90° . We first divide the plane into quadrants.

1. When the angle θ is in the second quadrant, i.e $90^\circ < \theta < 180^\circ$, to get the associated acute angle we subtract θ from 180° ($\alpha = 180^\circ - \theta$).
2. When the angle θ is in the third quadrant, i.e $180^\circ < \theta < 270^\circ$, to get the associated acute angle we subtract 180° from θ , ($\alpha = \theta - 180^\circ$).
3. When the angle θ is in the fourth quadrant, i.e $270^\circ < \theta < 360^\circ$, to get the associated acute angle we subtract θ from 360° ($\alpha = 360^\circ - \theta$).

The signs of the trigonometric ratios depend on the signs of x and y , the coordinates of R . In the first quadrant both x and y are positive, and since r is always positive, we have

$$\sin \theta = \frac{y}{r} > 0, \cos \theta = \frac{x}{r} > 0, \tan \theta = \frac{y}{x} > 0.$$

In the second quadrant x is negative and y is positive, so that we have

$$\sin \theta = \frac{y}{r} > 0, \cos \theta = \frac{x}{r} < 0, \tan \theta = \frac{y}{x} < 0.$$

In the third quadrant both x and y are negative so that

$$\sin \theta = \frac{y}{r} < 0, \cos \theta = \frac{x}{r} < 0, \tan \theta = \frac{y}{x} > 0.$$

Similarly in the fourth quadrant x is positive and y is negative so that

$$\sin \theta = \frac{y}{r} < 0, \cos \theta = \frac{x}{r} > 0, \tan \theta = \frac{y}{x} < 0.$$

The trigonometric ratio of an angle greater than 90° will be the ratio of its associated acute angle but with a positive or negative sign depending on whether the ratio, $\frac{y}{r}$, $\frac{x}{r}$ or $\frac{y}{x}$, is positive or negative in the quadrant where the angle falls.

Angles measured in the anticlockwise direction are positive while angles measured in clockwise direction are negative.

We have already seen that sine is negative in the fourth quadrant while cosine is positive there. We may conclude that for any angle θ ;

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = -\tan \theta.$$

Thus cosine is an even function while sine is an odd function.

It is useful to memorize the trigonometric ratios for the commonly occurring (standard) angles $30^\circ, 45^\circ, 60^\circ$, which can be obtained, conveniently, from the right-angled isosceles triangle and an equilateral triangle. The term 'cosine' is not accidental. Cosine of θ is the sine of $90^\circ - \theta$. If α and β are two angles, which sum up to 90° , then they are said to be complementary. β is the complement of α and vice versa. If α and β are complementary angles in a right-angled triangle, then we see that $\sin \alpha = \cos \beta = \cos(90^\circ - \alpha)$, $\cos \alpha = \sin \beta = \sin(90^\circ - \alpha)$, $\tan \alpha = \cot \beta = \cot(90^\circ - \alpha)$ and $\cot \alpha = \tan \beta = \tan(90^\circ - \alpha)$.

Example 5.2.1. Write down the exact value of each of the following:

1. $\sin 120^\circ$

2. $\cos \frac{\pi}{3}$

3. $\cot 225^\circ$

4. $\sec(-30)^\circ$

Example 5.2.2. Find all the values of θ in the range $0 \leq \theta \leq 2\pi$ such that

$$2 \sin \theta + \sqrt{3} = 0.$$

5.3 Trigonometric identities

The standard trigonometric identities are normally found on a formulae sheet (table). However, the key identities should actually be memo-

rised. In fact, you only have to remember one or two, from which the rest may then be derived. A sufficient minimal set to remember is in fact:

$$\cos^2 \theta + \sin^2 \theta = 1$$

and

$$\sin(A + B) = \sin A \cos B + \cos A \sin B.$$

From $\cos^2 \theta + \sin^2 \theta = 1$, we obtain $1 + \tan^2 \theta = \sec^2 \theta$ and $\cot^2 \theta + 1 = \csc^2 \theta$. These are called Pythagorean identities.

5.3.1 Compound angle formulae

The basic compound angle formulae are:

1. $\sin(A + B) = \sin A \cos B + \cos A \sin B$
2. $\sin(A - B) = \sin A \cos B - \cos A \sin B$
3. $\cos(A + B) = \cos A \cos B - \sin A \sin B$
4. $\cos(A - B) = \cos A \cos B + \sin A \sin B$
5. $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
6. $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

By putting $A = B$ in the compound angle identities, we immediately get the double angle formulae:

1. $\sin(2A) = 2 \sin A \cos A$
2. $\cos(2A) = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A$
3. $\tan(2A) = \frac{2 \tan A}{1 - \tan^2 A}$

To understand the compound formulae discussed above, we have the following example.

Example 5.3.2. Evaluate

1. $\sin 15^\circ$.

2. $\cos 75^\circ$

3. $\sin 105^\circ$

4. $\tan(-75^\circ)$

Example 5.3.3. Prove that :

1. $\cos^2 A - \sin^2 A \equiv 2 \cos^2 A - 1$

2. $\cot \theta + \tan \theta \equiv \csc \theta \sec \theta$

3. $\frac{1}{1+\sin \theta} + \frac{1}{1-\sin \theta} \equiv 2 \sec^2 \theta$

4. $\tan^2 \theta = \sin^2 \theta (1 + \tan^2 \theta)$.

5.4 Basic Trigonometric equations

5.4.1 Basic Trigonometric equations

A basic trig equation will $\sin \theta$, $\cos \theta$, $\tan \theta$ will usually have two solutions for $0^\circ \leq \theta \leq 360^\circ$. To solve a basic equation such as $\sin \theta = k$, we will require the following steps.

1. Find the first quadrant angle α for which $\sin \alpha = |k|$

2. Find the quadrants in which theta must lie

3. Determine the corresponding angles for those quadrants.

Example 5.4.2. Solve the following equation for $0^\circ \leq \theta \leq 360^\circ$:

1. $\tan \theta = -\frac{1}{\sqrt{3}}$.

2. $\cos \theta = \frac{\sqrt{2}}{2}$.

3. $\sin(\theta - 30^\circ) = -\frac{\sqrt{3}}{2}$.

5.4.3 Equations with more than one function

Further types of trigonometric equations can be solved by correctly applying the trigonometric identities we have looked at so far. The idea is to reduce the given equation to basic trig equations.

Example 5.4.4. Solve each of the following for $0^\circ \leq \theta \leq 360^\circ$:

1. $2 \cos^2 \theta + \cos \theta = 1.$

2. $2 \sin \theta = \tan \theta$

3. $\sqrt{3} \tan^2 \theta + 3 \tan \theta = \tan \theta + \sqrt{3}.$

5.4.5 Graphs of Trigonometric functions

Consider the graph of $y = \sin x$ for $0 \leq x \leq 2\pi$. The table below shows the corresponding values of $\sin x$ for given values of x .

x	0°	90°	180°	270°	360°
$y = \sin x$	0	1	0	-1	0
$y = \cos x$	1	0	-1	0	1

Plotting these values and connecting them yields the following curves.

The sine function will repeat itself after 2π radians, and so does the cosine function. Such functions or curves are said to be **periodic**. We can therefore expect the graph to repeat itself as x decreases to $-\infty$ and as x increases to ∞ . Using this fact we sketch the curve for $-2\pi \leq x \leq 2\pi$ as shown below.

When we graph such a periodic curve within its period of $[0, 2\pi]$ or $[0^\circ, 360^\circ]$, there are 5 points that help us in sketching the curve. We call them the **critical points** of the periodic function. By definition, the critical points are the points in which the curve either crosses the axis of symmetry, reached a maximum or minimum.

5.4.6 Amplitude and Period

Now that we know the shape and behavior of the sine and cosine curves, we will now do some things to alter the behavior. Our goal is to predict the shape of the curve without resorting to actually graphing it.

Example 5.4.7. Sketch each of the following graphs for $0 \leq x \leq 2\pi$:

1. $y = 3 \cos x$
2. $y = -2 \sin x$
3. $y = \frac{2}{3} \sin x$

Definition 5.4.8. The Amplitude of $y = a \sin x$ and $y = a \cos x$ is the maximum or largest value of y and is given by $|a|$. The range of the curve is $[-a, a]$.

1. We define the shape of the curve as follows:
 - (a) If $a > 0$, we obtain a normal curve.
 - (b) If $a < 0$, we obtain a reversed curve.
2. We define vertical change of the curve as follows:
 - (a) If Amplitude > 1 , we obtain a vertical stretch.
 - (b) If Amplitude < 1 , we obtain a vertical shrink.
 - (c) If Amplitude $= 1$, no vertical change.

To understand the period of a function we have the following example.

Example 5.4.9. Sketch each of the following graphs for $0 \leq x \leq 2\pi$:

1. $y = \sin x$
2. $y = \sin 2x$
3. $y = \sin 3x$

Notice that by changing the coefficient of the x – variable, we control the number of complete cycles of the curve we produce. In $y = \sin x$, when the coefficient of x is 1, we get one cycle within 360° . In $y = \sin 2x$, we get two cycle within 360° . Thus, also change the **period** of the curve, the number of degrees the curve takes to complete one cycle.

Definition 5.4.10. The period of $y = \sin bx$ or $y = \cos bx$ is the number of degrees (or radians) the curve takes to complete a cycle. Thus

$$\text{Period} = \frac{360^\circ}{b} \text{ or } \frac{2\pi}{b}.$$

If the period is less than 360° , we get a compression. If the period is greater than 360° , we get an elongation.

Example 5.4.11. Sketch each of the following graphs for $0 \leq x \leq 2\pi$:

1. $y = \sin x$
2. $y = 2 + \sin x$
3. $y = -1 + \sin x$

Note that by changing the constant that is added or subtracted to the basic sine or cosine curve, we affect how the graph is shifted up or down. This is the **vertical shift** or vertical **translation**.

The curve in the form

$$y = d \pm a \sin bx \text{ or } y = d \pm a \cos bx,$$

will shift the curve up or down based on the value of d .

If $d > 0$, the curve shifts d units upward. If $d < 0$, the curve shifts d units downward.

Example 5.4.12. Sketch each of the following graphs for $0 \leq x \leq 2\pi$:

1. $y = \sin x$
2. $y = \sin(x - 30^\circ)$
3. $y = \sin(x + 45^\circ)$

Note that by changing the constant that is added or subtracted within the parentheses, we shift the periodic curve left or right. This is called the **horizontal translation or phase shift**.

A curve in the form of $y = \sin(x + c)$ or $y = \cos(x + c)$ will shift the curve right or left based on the value of c .

If $c > 0$, the curve shifts left. If $c < 0$, the curve shifts right.

In particular, $y = \sin(x + 10^\circ)$ shifts the curve 10 degrees to the left, while $y = \sin(x - 10^\circ)$, shifts the curve 10 degrees to the right.

Example 5.4.13. Find the amplitude, period and phase shift, and hence sketch each of the following graphs for $0 \leq x \leq 2\pi$:

1. $y = -2 + \frac{1}{2} \sin(x - 30^\circ)$
2. $y = \cos(2x + \frac{\pi}{2})$